Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/08/2005
PI Name:	Egolfopoulos, Fokion Ph.D.		
Project Title:	Detailed Studies on the Structure and Dynamics of Reacting Dusty Flows at Normal- and Micro-Gravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	egolfopo@rcf.usc.edu	**Fax:**	FY 213 740-8071
PI Organization Type:	UNIVERSITY	**Phone:**	213 740-0480
Organization Name:	University of Southern California		
PI Address 1:	Department of Aerospace and Mechanical Engineering		
PI Address 2:			
PI Web Page:			
City:	Los Angeles	**State:**	CA
Zip Code:	90089-1453	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	06/01/2000	**End Date:**	04/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Wu, Ming-shin	**Contact Phone:**	216-433-3781
Contact Email:	Ming-shin.Wu@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Campbell, Charles S. (University of Southern California)		
Grant/Contract No.:	NCC3-834		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		
	The experiments include the use of a counterflow burner apparatus, equipped with several diagnostic instruments such as Laser Doppler Velocimetry (LDV), Digital Particle Image Velocimetry (DPIV), thermocouples, and real-time flame visualization, for the study of strained reacting dusty flows. The counterflow system provides an ideal environment for the systematic study of the effect of strain rate on the particle dynamic response. The numerical aspects of this program include the development and use of quasi-one-dimensional and fully axisymmetric, two-phase modeling of the counterflow, with the use of detailed description of molecular transport and chemical kinetics in both phases. Experiments are conducted at normal and micro-gravity by utilizing the KC-135 aircraft. To obtain qualitative and quantitative information about the ignition, propagation, structure, and extinction behavior of reacting dusty flows at normal and micro-gravity, and gain physical insight into the controlling physico-chemical mechanisms. The research		

Task Description:	will allow for the systematic study of the effects of gas phase stoichiometry, strain rate, particle type and size, thermophoresis, and gravity on the ignition and burning characteristics of reacting dusty flows. The results are expected to be of interest to solid fuel combustion applications, explosions, material synthesis, and LDV and DPIV gas flow velocity measurement techniques. The understanding of the behavior of solid particles in reacting environments is very important. Inert particles can be present in reacting gas phase mixtures whose burning intensity can be affected. From a safety point of view, it is very important to understand how inert particles both hot and cold, can affect the ignitability and flammability limits of reacting gases. The understanding of the behavior of reacting particles is also very important, given that solids can be used in numerous earth- and space-based combustion processes. There is increased interest in clean solid burning for power generation in the not-very-distant future. Solids are also the chief candidates for space propulsion including future spaceships that will fly to Mars and return, utilizing the Martian carbon-dioxide. Furthermore, accidents can occur when a reacting dust accumulates in air and, in the presence of an ignition source, can result in explosion. Such explosions can occur during lumber milling, in grain elevators, and in mine galleries. Gravity forces particles to settle out of a gas flow. At normal gravity both gravity and the particle inertia can cause a non-convergence of the particle and gas flow fields, as is the case for large particles and low velocity gas flows for which gravity may dominate the hydrodynamic effects. Performing the experiments in micro-gravity allows the gravitational settling to be eliminated leaving only the hydrodynamic effects. Comparison between the normal and micro-gravity results will allow gravitational effects to be understood.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Within the scope of this work, an experimental combustion apparatus was developed in laboratory consisting of opposed-jet and stagnation type burners. A particle-seeding unit, which would work both at 1-g and µg was developed and manufactured. An experimental combustion rig, similar to the laboratory one, was designed and built for experimentation in µg environment. In this way, the stagnation flow type system, which is considered as one of the best techniques to perform systematic study on flames was introduced, for the first time, to a relatively long µg period facility, in this case the KC-135. The extinction characteristics of dusty premixed and non-premixed systems were investigated experimentally in 1-g and µg. The extinction conditions of flames seeded with inert or combustible particles were investigated as a function of strain rate (flow rate), particle type, particle diameter, particle delivery rate to the flow, gas-phase chemical composition and temperature, flame thickness and orientation. A CHEMKIN-based quasi-1D stagnation flow flame numerical code incorporating the presence of particles was developed to advance the current understanding of the physics behind these flows. Numerical simulations were performed to simulate the 1-g and µg extinction experiments and the adequacy of available theoretical models were assessed. The effects of Stokes drag, gravity and thermophoresis on particles in typical reacting flows were studied. The underlying physics of the counter-intuitive differences between the extinction behaviors of flames in 1-g and µg were explained, and the effects of relatively small and large particles were addressed. It was demonstrated that meaningful scaling of interactions between the two phases is not possible due to complex couplings between thermal and dynamic parameters of two-phase flows. A numerical investigation on the ignition behavior of premixed and non-premixed flames by hot inert particles in opposed-jet configurations was conducted. The effects of reactant-composition, strain rate, orientation of particle seeding, particle injection temperature and velocity were demonstrated. The differences with the ignition achieved by the interaction of combustible gases with hot gases were identified. The chemical pathways were analyzed and sensitivity analysis was performed, both for the first time, to determine the dominant kinetic paths of this unique type of flows.
Bibliography Type:	Description: (Last Updated: 10/19/2009)
Articles in Peer-reviewed Journals	Andac MG, Egolfopoulos FN, Campbell CS, Lee JC. "Effects of combustible dust clouds on premixed flame extinction in normal- and micro-gravity." Proceedings of the Combustion Institute, 2005 Jan;30(2):2369-77. http://dx.doi.org/10.1016/j.proci.2004.08.256 , Jan-2005
Articles in Peer-reviewed Journals	Andac MG, Egolfopoulos FN, Campbell CS. "Hot-gas ignition of non-premixed methane flames in the presence of inert particles." Proceedings of the Combustion Institute, 2005 Jan;30(1):431-7. http://dx.doi.org/10.1016/j.proci.2004.08.053 , Jan-2005